Bistable Mechanism work

Introduction

A *bistable* mechanical system has two stable equilibrium positions separated by an energy barrier. If it is not disturbed, it stays in whichever well it already occupies. Moving from one well to the other requires enough input energy to cross the intervening saddle; the rapid transition is the *snap-through* event [[1]; [2]]. For the Rolly hook, the first well holds the hook retracted. A RobStride RS05 quasi-direct-drive motor drives the surrounding structure with a sinusoidal force or displacement. Near resonance, the hook motion grows until the relative coordinate crosses the saddle. After snap-through the second well holds the hook deployed without continuous motor power. The model here is therefore concerned with three quantities: the barrier height, the natural frequency about the retracted state, and the actuator amplitude needed to reach the saddle. The physical model derived and simulated here is **Simulation 2 (Physical)**: a compliant curved-beam bistable mechanism following the geometry of Qiu et al. [[1]] and the 3D-printed Design 9 of Zolfagharian et al. [[3]], with a companion two-degree-of-freedom tuned mass damper (TMD) model [[4]].

Variable Map

The MATLAB variable names are mapped to the symbols in this document below. This is also the convention used in the regenerated figures.

- `h_apex` $= h$: undeformed beam apex height.
- `t_beam` $= t$: beam thickness.
- `l_span` $= l$: horizontal span between beam end supports.
- `Q_beam` or `Q` $= Q = \frac{h}{t}$: bistability ratio. The scripts require $Q \geq 2.35$.
- `n_cells` $= N$: number of parallel bistable beams. Parallel beams share the same displacement and add force/stiffness.
- `d_stroke_1cell` $= d_{f,1}$: stroke of one beam.
- `d_stroke` $= d_f$: mechanism stroke used by the ODE. In the parallel model this equals `d_stroke_1cell`.
- `d_saddle` $= d_s$: unstable saddle displacement; snap-through occurs when x_{rel} crosses this value.
- `F_push_1cell` $= F_{\text{push,1}}$: peak force for one beam.
- `F_push_total` $= F_{\text{push,N}}$: total pack peak force, $NF_{\text{push,1}}$.
- `A_coeff` $= A$: cubic force coefficient in equation Equation 7.
- `k_eff_s1` $= k_{\text{eff,1}}$: small-signal stiffness about State 1.
- `m_hook` $= m_h$: hook/reference mass.
- `M_plate` $= M_P$: rigid plate, motor housing, and external-block mass.
- `xP` $= x_P$: absolute plate displacement.
- `xr` or `x_rel` $= x_{\text{rel}}$: hook displacement relative to the plate.
- `A_min_c2` $= A_{\text{min}}$: minimum prescribed plate amplitude for Configuration 2.

Compliant Curved-Beam Bistable Mechanism

Beam Geometry - Qiu Cosine Profile

Qiu et al. [[1]] proposed the initial (stress-free) shape of the curved beam as the first buckling mode of a fixed-end Euler beam, a cosine profile:

$$y_0(\xi) = \frac{h}{2}\left[1 - \cos\left(\frac{2\pi\xi}{l}\right)\right], 0 \leq \xi \leq l \tag{1}$$

where h is the apex height [m] and l is the horizontal span [m]. The beam thickness is t [m], and the material elastic modulus is E [Pa]. Key geometric parameters:

$$Q = \frac{h}{t} \text{ (apex-height-to-thickness ratio)} \tag{2}$$

$$I = \frac{t^3}{12} \text{ (second moment of area per unit width)} \tag{3}$$

Bistability Condition
For a curved beam to exhibit bistability, two design conditions matter most [[1]; [3]]:
1. **Q-ratio condition**: The apex height must substantially exceed

the beam thickness:

$$Q = \frac{h}{t} \geq 2.35 \tag{4}$$

Qiu et al. report that a second-mode-constrained curved beam becomes bistable once this geometric ratio is large enough; their analysis and force-displacement plots put the threshold at about $Q = 2.35$ [[1]]. The MATLAB model now enforces $Q \geq 2.35$ directly.
1. **Mode-2 suppression**: For a single beam, buckling mode 2 is

asymmetric and prevents robust bistability. In the Zolfagharian Design 9, a *double-curved beam* connected by a central membrane at mid-span suppresses mode 2, so the snap-through path follows the symmetric constrained-beam behavior described by Qiu et al. [[1]].

Force-Displacement Relationship

Strain Energy Formulation
Using the Euler-Bernoulli beam assumptions, let the initial shape be $y_0(\xi)$ and the transverse displacement from that shape be $w(\xi)$, so $y(\xi) = y_0(\xi) - w(\xi)$ where w is the transverse displacement relative to the initial shape. The bending strain energy is then [[1]]:

$$U_b = \frac{EI}{2}\int_0^l \left(\frac{\mathrm{d}^2 w}{\mathrm{d}\xi^2}\right)^2 \mathrm{d}\xi \tag{5}$$

The axial compressive strain energy (due to shortening of the beam's projected length) is:

$$U_a = \frac{EA}{2l}\left(\int_0^l \frac{\mathrm{d}y_0}{\mathrm{d}\xi}\frac{\mathrm{d}w}{\mathrm{d}\xi}\,\mathrm{d}\xi\right)^2 \tag{6}$$

where $A = t \cdot b$ is the cross-sectional area (for width b). The force-displacement relationship follows from virtual work: $F = \frac{\partial(U_b+U_a)}{\partial}d$, where d is the mid-point displacement (snapping direction).

Normalized Force-Displacement
Qiu et al. [[1]] show that the force-displacement curve depends only on Q and the normalized displacement $\bar{d} = \frac{d}{h}$. For the double-beam (mode-2 suppressed) configuration, the curve has the characteristic S-shape:

- **State 1** ($\bar{d} = 0$): zero force (equilibrium).
- **Peak push force** F_{push} at $\bar{d} \approx 0.27$:

maximum force required to initiate snap-through.

- **Saddle** ($\bar{d} = \bar{d}_s$): force returns to zero

(unstable equilibrium).

- **Valley force** $F_{\text{pop}} < 0$ at $\bar{d} \approx 0.82$:

mechanism pulls itself toward State 2.

- **State 2** ($\bar{d} = 1$): zero force (second equilibrium).

Cubic Polynomial Fit
The simulation uses a cubic fit for the applied force in the compression-test convention. This is not a replacement for Qiu's modal solution or Zolfagharian's FEA; it is a compact surrogate with the correct equilibrium points and peak force:

$$F_{\text{ext}}(d) = Ad(d - d_s)\left(d - d_f\right) \tag{7}$$

The coefficient A is calibrated so the force at the chosen peak location matches the Design 9 force level from Zolfagharian et al. [[3]]:

$$A = \frac{F_{\text{push}}}{d_{\text{push}}\left(d_{\text{push}} - d_s\right)\left(d_{\text{push}} - d_f\right)} \tag{8}$$

where $d_{\text{push}} \approx 0.27 d_f$ is the location of the force peak. The *internal restoring force* on an attached mass (Newton's third law) is:

$$F_{\text{restore}}(d) = -F_{\text{ext}}(d) = -Ad(d - d_s)\left(d - d_f\right) \tag{9}$$

Potential Energy (Curved-Beam Model)
Integrating the restoring force analytically:

$$V(d) = -\int_0^d F_{\text{restore}}(\xi)\,\mathrm{d}\xi = A\int_0^d \xi(\xi - d_s)\left(\xi - d_f\right)\mathrm{d}\xi \tag{10}$$

$$= A\int_0^d \left[\xi^3 - \left(d_s + d_f\right)\xi^2 + d_s d_f \xi\right]\mathrm{d}\xi \tag{11}$$

$$= A\left[\frac{d^4}{4} - \frac{\left(d_s + d_f\right)}{3}d^3 + \frac{d_s d_f}{2}d^2\right] \tag{12}$$

The energy barrier from State 1 to the saddle is:

$$\Delta E_1 = V(d_s) - V(0) = A\left[\frac{d_s^4}{4} - \frac{\left(d_s + d_f\right)}{3} d_s^3 + \frac{d_s d_f}{2} d_s^2\right] \tag{13}$$

Natural Frequencies at Each Stable State
Linearizing $F_{\text{restore}}(d)$ about each equilibrium: **State 1** ($d = 0$):

$$k_{\text{eff},1} = . - \frac{\mathrm{d}F_{\text{restore}}}{\mathrm{d}d}|_{\text{d=0}} = A d_s d_f \tag{14}$$

$$\omega_{\text{n},1} = \sqrt{\frac{k_{\text{eff},1}}{m}}, f_{\text{n},1} = \frac{1}{2\pi}\sqrt{\frac{A d_s d_f}{m}} \tag{15}$$

State 2 ($d = d_f$):

$$k_{\text{eff},2} = . - \frac{\mathrm{d}F_{\text{restore}}}{\mathrm{d}d}|_{\text{d=d\_f}} = A d_f \left(d_f - d_s\right) \tag{16}$$

$$\omega_{\text{n},2} = \sqrt{\frac{k_{\text{eff},2}}{m}}, f_{\text{n},2} = \frac{1}{2\pi}\sqrt{\frac{A d_f \left(d_f - d_s\right)}{m}} \tag{17}$$

Since $d_s < d_f$ by definition, and typically $d_s \approx 0.55 d_f$: $k_{\text{eff},2} = A d_{f\left(d_f - d_s\right)} \approx 0.45 A d_f^2$, while $k_{\text{eff},1} = A d_s d_f \approx 0.55 A d_f^2$, so $k_{\text{eff},2} < k_{\text{eff},1}$ and thus $f_{\text{n},2} < f_{\text{n},1}$.

Parallel Beam-Pack Mechanics
The current robot layout is treated as a parallel pack of identical curved beams. Each beam sees the same apex displacement, so the travel does not add:

$$d_{\text{f,N}} = d_{\text{f,1}} \text{ (mechanism stroke)} \tag{18}$$

$$d_{\text{s,N}} = d_{\text{s,1}} \text{ (saddle position)} \tag{19}$$

The forces add across the pack:

$$F_{\text{push,N}} = N F_{\text{push,1}} \tag{20}$$

The cubic coefficient therefore scales linearly:

$$A_N = N A_1 \tag{21}$$

and the State-1 stiffness becomes:

$$k_{\text{eff,1,N}} = A_N d_s d_f = N k_{\text{eff,1,1}} \tag{22}$$

Thus adding parallel beams raises stiffness, stored energy, and natural frequency:

$$f_{\text{n,1,N}} = \sqrt{N} f_{\text{n,1,1}} \tag{23}$$

This is the source of the earlier displacement error: a series-cell assumption multiplied the stroke and saddle travel, while the physical beam pack should share displacement and add force.

1-DOF Hook Snap-Through Model
The hook mass m is connected to the fixed robot body through the bistable mechanism. External vibration $F_{\mathrm{exc}}(t)$ is transmitted through the robot structure. The equation of motion is:

$$m\ddot{d} + c\dot{d} + F_{\mathrm{restore}}(d) = F_{\mathrm{exc}}(t) \tag{24}$$

where $c = 2\zeta m\omega_{\mathrm{n},1}$. The excitation is a linear chirp (frequency-modulated sinusoid):

$$F_{\mathrm{exc}}(t) = F_0 \sin\left[2\pi\left(f_{\mathrm{lo}}t + \frac{f_{\mathrm{hi}} - f_{\mathrm{lo}}}{2t_f}t^2\right)\right] \tag{25}$$

where f_{lo}, f_{hi} are the chirp bounds [Hz] and t_f is the simulation duration [s]. As the instantaneous frequency passes through $f_{\mathrm{n},1}$, resonance amplifies the displacement until snap-through.

2-DOF Tuned Mass Damper Model
Following Zolfagharian et al. [[3]] (their Figure 4), the main structure mass m_1 is supported by spring k_{main} and driven harmonically. The TMD mass m_2 is attached to the same flexible ruler at distance L_2 from the fixed end, where the bistable mechanism repositions it. The coupled equations of motion are:

$$m_1\ddot{x}_1 + c_{\mathrm{main}}\dot{x}_1 + k_{\mathrm{main}}x_1 + k_{b(x_1 - x_2)} + c_{b(\dot{x}_1 - \dot{x}_2)} = F_0\sin(\omega t) \tag{26}$$

$$m_2\ddot{x}_2 - k_{b(x_1 - x_2)} - c_{b(\dot{x}_1 - \dot{x}_2)} = 0 \tag{27}$$

where k_b and c_b are the stiffness and damping of the bistable TMD connection, linearized at the current stable state.

Stiffness of the Flexible Ruler

The effective spring stiffness of a flexible cantilever beam (ruler) at distance L from the fixed support is [[3]]:

$$k = \frac{3EI_r}{L^3} \tag{28}$$

where E_r is the ruler's elastic modulus, I_r is its second moment of area. When the bistable mechanism shifts the TMD mass from L_1 to $L_2 = L_1 + d_f$:

$$k_{\mathrm{TMD},2} = \frac{3E_rI_r}{L_2^3} = \frac{3E_rI_r}{\left(L_1 + d_f\right)^3} < k_{\mathrm{TMD},1} \tag{29}$$

This reduction in TMD stiffness shifts the TMD natural frequency downward, changing the frequency at which it suppresses main-mass vibration.

Steady-State Frequency Response

In the frequency domain, the steady-state response of x_1 to $F_0e^{i\omega t}$ is $X_1e^{i\omega t}$. The impedance formulation gives:

$$((Z_{11}, Z_{12})\ \ (Z_{21}, Z_{22}))((X_1)\ \ (X_2)) = ((F_0)\ \ (0)) \tag{30}$$

where

$$Z_{11} = (k_{\text{main}} + k_b - m_1\omega^2) + i\omega(c_{\text{main}} + c_b) \tag{31}$$

$$Z_{12} = Z_{21} = -(k_b + i\omega c_b) \tag{32}$$

$$Z_{22} = (k_b - m_2\omega^2) + i\omega c_b \tag{33}$$

Solving by Cramer's rule:

$$X_1 = \frac{F_0 Z_{22}}{Z_{11}Z_{22} - Z_{12}^2} \tag{34}$$

The amplitude $|X_1(\omega)|$ vs. $\frac{\omega}{2\pi}$ [Hz] is plotted for three configurations: no TMD, TMD in State 1, TMD in State 2.

Base Excitation via Motor Torque Oscillation Through Shared Structure

Physical Setup

Mechanism component chain

The Zolfagharian Design 9 bistable mechanism consists of the following rigid bodies connected in series:

1. **Robot chassis / ground** - the fixed reference frame.
2. **Spring isolators** (k_s, c_s) - soft mounting elements

(rubber mounts, vibration isolators) between the chassis and the plate assembly.

1. **Rigid plate assembly** (M_P) - one rigid body comprising the

plate, the motor housing, and the *external blocks*. The external blocks are the side walls to which the bistable beam ends are pinned.

1. **Bistable curved beams** - the compliant Qiu cosine-profile beams

spanning between the external blocks. These provide the nonlinear restoring force $F_{\text{restore}}(x_{\text{rel}})$ and potential energy $V(x_{\text{rel}})$.

1. **Hook** (m_h, also called the *reference mass*) - the

deployable element attached at the beam apex. In State 1 the beam is arched and the hook is retracted; in State 2 the beam has snapped through and the hook is deployed.

The hook is connected to the plate assembly *only* through the bistable beams - there is no rigid link. When the plate assembly vibrates, the hook cannot follow instantaneously; the resulting **relative displacement** x_{rel} between the beam apex (hook) and the beam ends (external blocks on plate) is the quantity that drives snap-through.

```
ROBOT CHASSIS
      |
      |  k_s, c_s   <- spring isolators / rubber feet
      |
RIGID ASSEMBLY  (M_P = motor + plate + external blocks)
      |                  external blocks = beam end pinned supports
      |
    BEAM      BEAM    <- bistable curved beams (Qiu cosine profile)
      |          |
      +----+-----+
           |
         HOOK  (m_h = reference mass at beam apex)
```

Listing 1: Physical component chain for the Zolfagharian bistable deployment mechanism.

Two excitation configurations

Configuration 1 - Motor on plate assembly.

The motor housing is bolted directly to the plate (part of M_P). The motor shaft drives an eccentric mass or proof mass whose reaction force enters the plate through the bearing. The plate assembly vibrates; the hook responds via base excitation through the beams.

$$\text{Motor} \to \text{Plate} + \text{External blocks} \to \text{Beams} \to \text{Hook} \tag{35}$$

Configuration 2 - Motor off plate, drives via bearing.

The motor housing is *not* part of the plate assembly. The motor output shaft drives the plate+external-blocks assembly directly through a bearing coupling (lever arm, crank, or shaft). In this configuration the motor can *prescribe* the plate displacement:

$$x_{P(t)} = A \sin\left(\omega_{\mathrm{n},1} t\right) \tag{36}$$

The hook responds to this base excitation through the beams. The motor housing mass is excluded from M_P.

$$\text{Motor} \to \text{Bearing} \to \text{Plate} + \text{External blocks} \to \text{Beams} \to \text{Hook} \tag{37}$$

Configuration 2 avoids the tuned-mass-damper anti-resonance problem because the motor enforces the plate amplitude regardless of the hook's dynamics.

Coordinates

Two coordinate systems are used throughout:
- $x_{P(t)}$: absolute displacement of the rigid plate assembly [m]

(positive in the snap direction, i.e. from State 1 toward State 2)
- $x_{H(t)}$: absolute hook displacement [m]
- $x_{\mathrm{rel}}(t) \equiv x_{H(t)} - x_{P(t)}$: hook displacement

relative to the plate assembly [m] - this is the quantity that enters the bistable restoring force and potential energy

Snap-through occurs when x_{rel} crosses the saddle point d_s with positive velocity.

Motor Torque Excitation Force

In torque/current control mode the RS05 applies a sinusoidal output torque $\tau_{\text{out}}(t)$. Through a coupling of moment arm r_{arm} [m] this becomes a linear force on the plate:

$$F_{\text{motor}}(t) = \frac{\tau_{\text{out}}}{r_{\text{arm}}} \sin(\omega_d t) \equiv F_0 \sin(\omega_d t) \tag{38}$$

where τ_{out} is the **measured output torque** read from the motor's torque-speed (T-N) curve at the instantaneous operating condition, and ω_d is the drive frequency set by the controller command. The amplitude F_0 is therefore *independent of frequency* - a key difference from a rotating-unbalance source whose amplitude grew as ω^2. For a linear chirp sweep from f_{lo} to f_{hi} over total time t_f, the instantaneous frequency and accumulated phase are:

$$f_{\text{inst}}(t) = f_{\text{lo}} + \frac{f_{\text{hi}} - f_{\text{lo}}}{t_f} t \tag{39}$$

$$\varphi(t) = 2\pi \left[f_{\text{lo}} t + \frac{f_{\text{hi}} - f_{\text{lo}}}{2t_f} t^2 \right] \tag{40}$$

The torque-driven force during a chirp is:

$$F_{\text{motor}}(t) = F_0 \sin(\varphi(t)) \tag{41}$$

The amplitude envelope F_0 is constant throughout the sweep, so the force delivered at $f_{\text{n,1}}$ equals the force at any other frequency in the band - the mechanism receives full excitation amplitude at every point in the chirp.

Gearbox Effects on Vibration Transmission

Any gearbox in the drivetrain between the motor rotor and the bistable plate acts as a mechanical filter. The gear ratio n_g amplifies the rotor's reflected inertia and creates a low-pass cut-off frequency above which vibrations are attenuated before reaching the plate. Two architectures relevant to bistable excitation are treated below.

Planetary Gearboxes

A planetary gearbox achieves its ratio n_g through a sun-planet-ring arrangement. Typical ratios range from 3:1 to about 20:1 per stage; multi-stage units reach 100:1. QDD motors use a single low-ratio planetary stage ($n_g \leq 10$) specifically to preserve backdrivability and broadband force transmission. **Reflected rotor inertia.** At the output shaft, the effective inertia is:

$$J_{\text{eff}} = J_{\text{rotor}} n_g^2 + J_{\text{gears}} + J_{\text{output}} \tag{42}$$

The n_g^2 factor dominates: even a modest 5:1 ratio makes the rotor inertia appear $25 \times$ larger at the output. Combined with the finite gear-mesh stiffness k_{gear}, this creates a mechanical low-pass cut-off frequency:

$$f_{\mathrm{cut}} \approx \frac{1}{2\pi}\sqrt{\frac{k_{\mathrm{gear}}}{J_{\mathrm{eff}}}} \tag{43}$$

Vibrations at frequencies above f_{cut} are attenuated before reaching the plate; those well below f_{cut} pass through with little attenuation. A low gear ratio keeps f_{cut} high, so the motor remains substantially backdrivable and transmits a broad vibration band - a favorable property for this application. **Design rule:** verify that $f_{\mathrm{n},1} < f_{\mathrm{cut}}$ so that the bistable resonance lies in the pass-band of the actuator's mechanical filter. This check should use the actual mounted motor and linkage geometry, not the bare motor datasheet value. **Gear-mesh frequency.** The planet-gear tooth contacts produce a secondary vibration at:

$$f_{\mathrm{mesh}} = n_g f_{\mathrm{rotor}} N_{\mathrm{teeth}} \tag{44}$$

where N_{teeth} is the planet tooth count. If f_{mesh} happens to coincide with $f_{\mathrm{n},1}$, this harmonic can contribute beneficially to snap-through excitation without any additional control effort. A low single-stage ratio keeps f_{mesh} well above the bistable resonance band, avoiding unintended parametric excitation.

High-Ratio Straight-Tooth Gearboxes

Multi-stage spur or helical gearboxes achieve ratios of $n_g = 20{:}1$ to $n_g > 100{:}1$ by cascading two or more stages of spur (straight-tooth) or helical gears. They are inexpensive and compact relative to harmonic or cycloidal alternatives at the same ratio, but carry two significant penalties for bistable vibration excitation. **Reflected inertia penalty.** Because n_g is large, the n_g^2 term in equation Equation 42 dominates overwhelmingly. The cut-off frequency f_{cut} drops accordingly:

$$f_{\mathrm{cut}} \propto \frac{1}{n_g}\left(J_{\mathrm{eff}} \approx J_{\mathrm{rotor}} n_g^2\right) \tag{45}$$

At $n_g = 50$, a motor with $f_{\mathrm{cut}} = 500$ Hz at $n_g = 1$ drops to roughly 10 Hz - potentially *below* the bistable resonance. Vibrations above this cut-off are mechanically filtered; the plate never receives the high-frequency force content needed for resonant snap-through. **Backlash dead-band.** Multi-stage spur gearboxes accumulate backlash θ_{bl} (typically 5-30 arc-min total) from each mesh. Through the coupling lever arm r this creates a linear dead-band:

$$\delta_{\mathrm{bl}} = \theta_{\mathrm{bl}} r \tag{46}$$

During each oscillation half-cycle the drive must traverse δ_{bl} before force transmission resumes. The effective plate amplitude is reduced to:

$$A_{\mathrm{eff}} = A_{\mathrm{plate}} - \delta_{\mathrm{bl}} \tag{47}$$

For snap-through, the minimum plate amplitude from equation Equation 105 must be met *net of backlash*: $A_{\mathrm{eff}} \geq A_{\mathrm{min}}$, i.e.
$A_{\mathrm{plate}} \geq 2\zeta_h d_s + \delta_{\mathrm{bl}}$. If δ_{bl} is comparable to d_s, this constraint can become severe. **Gear-mesh noise.** Multi-stage meshes generate harmonics at integer multiples of f_{mesh} (equation Equation 44). If any harmonic coincides with $f_{\mathrm{n},1}$, it can excite unintended snap-through; this

is harder to avoid in a multi-stage box than in a single-stage planetary. **Design rule.** High-ratio straight-tooth gearboxes are suitable for bistable excitation only when $f_{\mathrm{n},1} \ll f_{\mathrm{cut}}(n_g)$, the accumulated backlash $\delta_{\mathrm{bl}} \ll d_s$, and the drive frequency is well below the first gear-mesh harmonic. In practice, these conditions are difficult to satisfy simultaneously at gear ratios above $\sim$ 20:1; a low-ratio QDD or cycloidal stage is strongly preferred.

Two-Degree-of-Freedom Equations of Motion

The plate is modelled as a rigid body on a linear elastic support (stiffness k_s, damping c_s). The hook mass m_h is connected to the plate through the bistable element (restoring force $F_{\mathrm{restore}}(x_{\mathrm{rel}})$, equation Equation 9) and a viscous damper c_h.

Plate equation of motion:

$$M_P \ddot{x}_P = F_{\mathrm{motor}}(t) \underbrace{- k_s x_P - c_s \dot{x}_P}_{\text{support restraint}} \underbrace{- F_{\mathrm{restore}}(x_{\mathrm{rel}}) + c_h \dot{x}_{\mathrm{rel}}}_{\text{reaction from hook}} \tag{48}$$

Hook equation of motion (absolute frame):

$$m_h \ddot{x}_H = F_{\mathrm{restore}}(x_{\mathrm{rel}}) - c_h \dot{x}_{\mathrm{rel}} \tag{49}$$

The sign convention in equation Equation 48 reflects Newton's third law: the bistable spring pulls the plate with force $-F_{\mathrm{restore}}$ (equal and opposite to what it exerts on the hook), and the dashpot exerts $+c_h \dot{x}_{\mathrm{rel}}$ on the plate.

Relative-Coordinate Formulation

Subtracting equation Equation 48 (divided by M_P) from equation Equation 49 (divided by m_h):

$$\ddot{x}_{\mathrm{rel}} = \ddot{x}_H - \ddot{x}_P = \underbrace{\frac{F_{\mathrm{restore}} - c_h \dot{x}_{\mathrm{rel}}}{m_h}}_{\text{dot.double(x)\_H}} - \underbrace{\frac{F_{\mathrm{motor}} - k_s x_P - c_s \dot{x}_P - F_{\mathrm{restore}} + c_h \dot{x}_{\mathrm{rel}}}{M_P}}_{\text{dot.double(x)\_P}} \tag{50}$$

This is the governing equation for the bistable snap. The full state vector is $\boldsymbol{y} = [x_P, \dot{x}_P, x_{\mathrm{rel}}, \dot{x}_{\mathrm{rel}}]^{\rightarrow} p$. The support parameters are defined from the plate natural frequency f_P:

$$k_s = M_P (2\pi f_P)^2 \tag{51}$$

$$c_s = 2\zeta_P M_P (2\pi f_P) \tag{52}$$

where ζ_P is the plate structural damping ratio.

Plate Transfer Function (Decoupled Analysis)

For small x_{rel} (early in the simulation, well before snap), the back-reaction of the bistable on the plate is negligible relative to the dominant terms F_{motor}, $k_s x_P$, and $c_s \dot{x}_P$. Under this approximation, the plate behaves as an independent SDOF oscillator:

$$M_P \ddot{x}_P + c_s \dot{x}_P + k_s x_P \approx F_{\mathrm{motor}}(t) \tag{53}$$

Assuming harmonic excitation at frequency ω with constant amplitude F_0, the steady-state plate displacement amplitude is:

$$|X_{P(\omega)}| = \frac{F_0}{\sqrt{\left(k_s - M_P\omega^2\right)^2 + \left(c_s\omega\right)^2}} \tag{54}$$

Substituting equation Equation 51 and equation Equation 52 and defining the frequency ratio $r_P = \frac{\omega}{\omega_P}$ where $\omega_P = 2\pi f_P$:

$$|X_{P(\omega)}| = \frac{\frac{F_0}{M_P\omega_P^2}}{\sqrt{\left(1 - r_P^2\right)^2 + \left(2\zeta_P r_P\right)^2}} = \frac{\frac{F_0}{k_s}}{\sqrt{\left(1 - r_P^2\right)^2 + \left(2\zeta_P r_P\right)^2}} \tag{55}$$

This is the standard frequency-response function of a driven SDOF oscillator. The amplitude peaks sharply at $r_P = 1$ (plate resonance, $\omega = \omega_P$), with the peak width governed by ζ_P.

Effective Excitation Force on the Bistable Hook

In the relative-coordinate frame, the plate's acceleration acts as an inertial (fictitious) force on the hook. The effective force amplitude at frequency ω is:

$$F_{\text{eff}}(\omega) = m_h\omega^2 \, |X_{P(\omega)}| \tag{56}$$

Substituting equation Equation 54:

$$F_{\text{eff}}(\omega) = \frac{m_h F_0 \omega^2}{\sqrt{\left(k_s - M_P\omega^2\right)^2 + \left(c_s\omega\right)^2}} \tag{57}$$

This is the force that the base excitation delivers to the bistable hook at frequency ω. Several observations follow directly:

1. The numerator scales as ω^2 (not ω^4 as in a

rotating-unbalance source): the force still grows with frequency, but more gradually. At low frequencies the inertial coupling is weak; at high frequencies the hook is increasingly driven by the plate's acceleration.

1. The denominator contains the plate resonance denominator. When ω

approaches ω_P, the denominator passes through a minimum, and F_{eff} passes through a maximum.

1. The maximum of F_{eff} occurs near (but not exactly at)

ω_P. If ω_P is designed to coincide with $\omega_{\text{n},1}$, the maximum excitation force is delivered precisely when the bistable hook is at its resonance - a double-resonance condition that amplifies the effective force by $Q_P \cdot Q_h$ relative to the quasi-static case.

General Snap-Through Condition

From equation Equation 101 (Optimization section), the minimum force required for snap-through at resonance $\omega = \omega_{\text{n},1}$ is:

$$F_{0,\min} = 2\zeta_h k_{\text{eff},1} d_s = 2\zeta_h m_h \omega_{\text{n},1}^2 d_s \tag{58}$$

Setting $F_{\text{eff}}(\omega_{\text{n},1}) \geq F_{0,\min}$ and substituting equation Equation 57:

$$\frac{m_h F_0 \omega_{\text{n},1}^2}{\sqrt{\left(k_s - M_P \omega_{\text{n},1}^2\right)^2 + \left(c_s \omega_{\text{n},1}\right)^2}} \geq 2\zeta_h m_h \omega_{\text{n},1}^2 d_s \tag{59}$$

Dividing both sides by $m_h \omega_{\text{n},1}^2$ (positive):

$$\frac{F_0}{\sqrt{\left(k_s - M_P \omega_{\text{n},1}^2\right)^2 + \left(c_s \omega_{\text{n},1}\right)^2}} \geq 2\zeta_h d_s \tag{60}$$

Rearranging, the **general snap condition** on the motor force amplitude is:

$$F_0 \geq 2\zeta_h d_s \sqrt{\left(k_s - M_P \omega_{\text{n},1}^2\right)^2 + \left(c_s \omega_{\text{n},1}\right)^2} \tag{61}$$

The right-hand side is entirely determined by known system parameters. Notably, m_h has cancelled: the snap condition is *independent of hook mass*. However, $\omega_{\text{n},1}$ itself depends on m_h through equation Equation 15, so the dependence is implicit.

Case 1: Tuned Plate ($f_P = f_{\text{n},1}$)

Derivation

Set the plate support natural frequency equal to the bistable natural frequency:

$$f_P = f_{\text{n},1} \Leftrightarrow \omega_P = \omega_{\text{n},1} \Leftrightarrow k_s = M_P \omega_{\text{n},1}^2 \tag{62}$$

Substituting into equation Equation 60. The term $k_s - M_P \omega_{\text{n},1}^2$ vanishes exactly, leaving only the damping term in the denominator:

$$\sqrt{\left(M_P \omega_{\text{n},1}^2 - M_P \omega_{\text{n},1}^2\right)^2 + \left(c_s \omega_{\text{n},1}\right)^2} = c_s \omega_{\text{n},1} \tag{63}$$

Substituting $c_s = 2\zeta_P M_P \omega_P = 2\zeta_P M_P \omega_{\text{n},1}$:

$$c_s \omega_{\text{n},1} = 2\zeta_P M_P \omega_{\text{n},1}^2 \tag{64}$$

The snap condition equation Equation 60 becomes:

$$\frac{F_0}{2\zeta_P M_P \omega_{\text{n},1}^2} \geq 2\zeta_h d_s \tag{65}$$

$$F_0 \geq 4\zeta_h \zeta_P M_P \omega_{\text{n},1}^2 d_s \text{ (Case 1: tuned plate)} \tag{66}$$

Physical Interpretation

Two points are useful for design:

1. **Scales with frequency.** Unlike a rotating-unbalance source,

$F_0 \propto \omega_{n,1}^2$: a higher bistable resonance requires more actuator force. The available force should be verified against this threshold after any change to geometry or mass, using the actuator's rated output at the chosen coupling arm length.

1. **Doubly discounted by damping.** The factor

$4\zeta_h\zeta_P$ means the threshold is $Q_P = \frac{1}{2\zeta_P}$ times smaller than the off-resonance quasi-static baseline $2\zeta_h M_P \omega_{n,1}^2 d_s$. This Q_P factor is the mechanical quality factor of the plate resonance. *Physical explanation:* at plate resonance the plate amplitude is amplified by Q_P relative to its quasi-static response; the hook then experiences Q_P times more inertial excitation, so Q_P times less actuator force is required.

Plate Amplitude at Snap Threshold

At the tuned threshold, from equation Equation 54 with the denominator reduced to $c_s\omega_{n,1}$:

$$|X_P|_{\text{tuned}} = \frac{F_0}{c_s\omega_{n,1}} = \frac{4\zeta_h\zeta_P M_P\omega_{n,1}^2 d_s}{2\zeta_P M_P \omega_{n,1}^2} = 2\zeta_h d_s \tag{67}$$

The plate displacement at threshold is $2\zeta_h d_s$. The hook displacement, amplified by $Q_h = \frac{1}{2\zeta_h}$ at bistable resonance, reaches d_s:

$$|x_{\text{rel}}|_{\text{at snap}} = Q_h\ |X_P| = \frac{2\zeta_h d_s}{2\zeta_h} = d_s \tag{68}$$

as expected. The two resonances (plate at f_P, hook at $f_{n,1}$) in series provide amplification $Q_P \cdot Q_h = \frac{1}{4\zeta_P\zeta_h}$ relative to the quasi-static case.

Case 2: Stiffness-Controlled Plate ($f_P > f_{n,1}$)

Regime Identification

When the plate support is stiffer than needed for resonance, $f_P > f_{n,1}$, the plate operates in the *stiffness-controlled* regime at the bistable's resonant frequency. Define the detuning ratio:

$$n \equiv \frac{f_P}{f_{n,1}} = \frac{\omega_P}{\omega_{n,1}} > 1 \tag{69}$$

In this regime $k_s > M_P\omega_{n,1}^2$, so the term $k_s - M_P\omega_{n,1}^2 = M_{P\left(\omega_P^2 - \omega_{n,1}^2\right)} > 0$ is the dominant contribution to the denominator of equation Equation 60.

Derivation

Substituting $k_s = M_P\omega_P^2 = M_P n^2 \omega_{n,1}^2$ and $c_s\omega_{n,1} = 2\zeta_P M_P \omega_P \omega_{n,1} = 2\zeta_P M_P n \omega_{n,1}^2$ into the denominator of equation Equation 61:

$$\sqrt{\left(k_s - M_P\omega_{n,1}^2\right)^2 + \left(c_s\omega_{n,1}\right)^2} \tag{70}$$

$$= \sqrt{M_P^2\omega_{n,1}^4(n^2-1)^2 + M_P^2\omega_{n,1}^4(2\zeta_P n)^2} \tag{71}$$

$$= M_P\omega_{\mathrm{n},1}^2\sqrt{\left(n^2-1\right)^2+\left(2\zeta_P n\right)^2} \tag{72}$$

Substituting into equation Equation 61:

$$F_0 \geq 2\zeta_h M_P\omega_{\mathrm{n},1}^2 d_s\sqrt{\left(n^2-1\right)^2+\left(2\zeta_P n\right)^2}\ (\text{Case 2: } n=\frac{f_P}{f_{\mathrm{n},1}}>1) \tag{73}$$

For light structural damping ($\zeta_P \ll 1$, so $2\zeta_P n \ll n^2-1$):

$$F_0 \approx 2\zeta_h M_P\omega_{\mathrm{n},1}^2 d_s(n^2-1)(n>1,\zeta_P \ll 1) \tag{74}$$

The threshold grows as (n^2-1), which is approximately n^2 for large n. A plate resonance twice as high as the bistable resonance ($n=2$) requires approximately $3\times$ the off-resonance inertia floor, and $75\times$ the tuned threshold.

Verification: Case 2 Reduces to Case 1 at $n=1$

Setting $n=1$ in equation Equation 73:

$$F_0 \geq 2\zeta_h M_P\omega_{\mathrm{n},1}^2 d_s\sqrt{0+\left(2\zeta_P\right)^2}=4\zeta_h\zeta_P M_P\omega_{\mathrm{n},1}^2 d_s \tag{75}$$

which recovers equation Equation 66.

Closed-Form Case Comparison

Closed-Form Comparison

Define the off-resonance baseline $T_0=2\zeta_h M_P\omega_{\mathrm{n},1}^2 d_s$. Then:

$$\text{Case 1 (tuned): } F_0 \geq 4\zeta_h\zeta_P M_P\omega_{\mathrm{n},1}^2 d_s = 2\zeta_P T_0 \tag{76}$$

$$\text{Case 2 (stiffness-controlled): } F_0 \geq T_0\sqrt{\left(n^2-1\right)^2+\left(2\zeta_P n\right)^2} \tag{77}$$

The ratio of Case 2 to Case 1 thresholds is unchanged from the rotating-unbalance analysis:

$$\frac{\text{Case 2 threshold}}{\text{Case 1 threshold}}=\frac{\sqrt{\left(n^2-1\right)^2+\left(2\zeta_P n\right)^2}}{2\zeta_P}\approx\frac{n^2-1}{2\zeta_P}(n>1,\zeta_P \ll 1) \tag{78}$$

For $\zeta_P=0.02$ and $n=2$: ratio $\approx \frac{4-1}{0.04}=75$. For $n=1.5$: ratio $\approx \frac{2.25-1}{0.04}=31$. For $n=1.1$: ratio $\approx \frac{1.21-1}{0.04}=5.3$. Even a modest 10% detuning ($n=1.1$) increases the required motor force by more than $5\times$ compared to the tuned case.

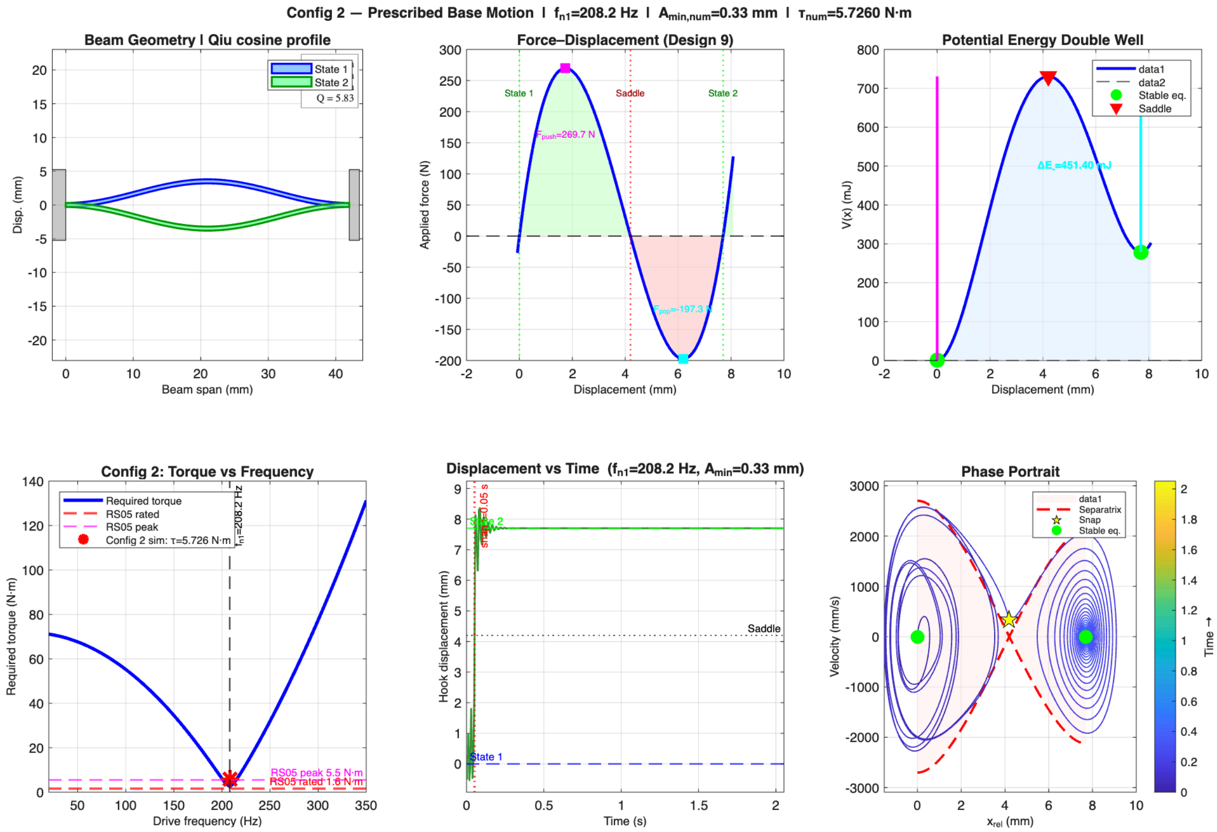

Figure 1: Time-domain snap-through simulation for Configuration 2 (prescribed base motion). The plate displacement $x_{P(t)}$ and relative hook displacement $x_{\mathrm{rel}}(t)$ are plotted; the hook crosses the saddle d_s and latches into State 2 without further motor input.

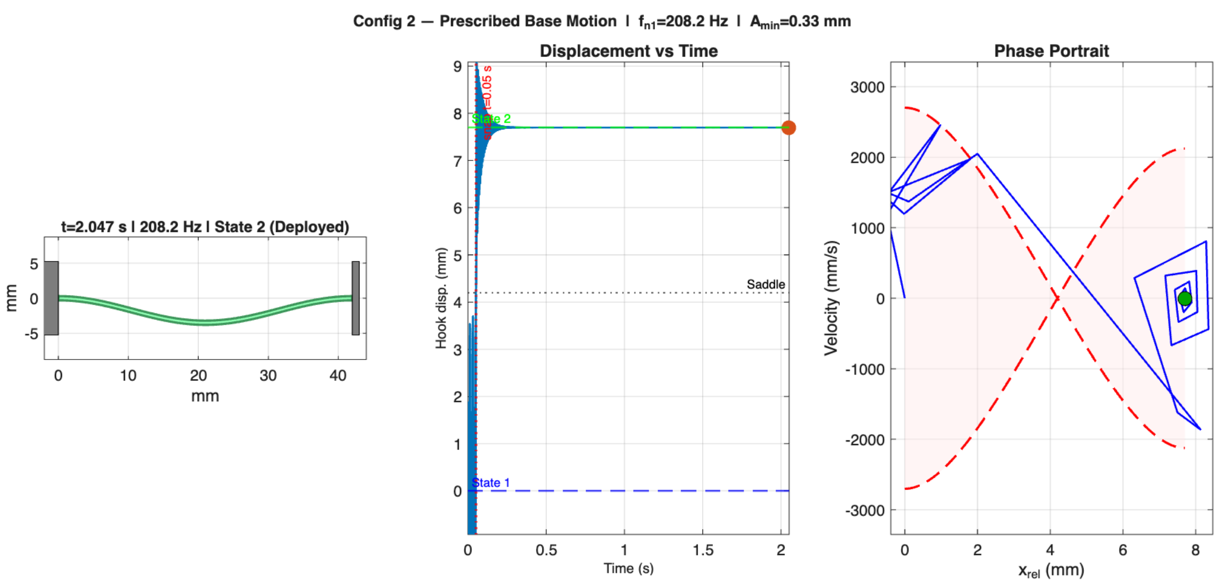

Figure 2: Animation snapshot of the Configuration 2 bistable excitation (prescribed base motion). The plate oscillates at $f_{\mathrm{n},1}$ and the hook builds resonant amplitude until snap-through occurs.

Motor Mounted on the Bistable Plate: Coupling Model and Assumptions

This subsection addresses the specific physical configuration assumed throughout the base-excitation model: the motor body is bolted directly to the same rigid plate that carries the bistable mechanism. It tracks the force path from an eccentric mass to the plate, shows why shaft angle is not a state variable in the hook dynamics, and states the assumptions explicitly.

Force Transmission Chain

When the motor shaft rotates, the eccentric mass m_e at radius e generates a centrifugal force. Before this force reaches the plate it passes through three mechanical interfaces, each with its own compliance:

1. **Shaft** (stiffness k_{sh}, damping c_{sh}):

The shaft transmits the eccentric mass force to the inner bearing race. A rigid shaft ($k_{\text{sh}} \to \infty$) transmits 100%; a flexible shaft at speeds near its *critical speed* $\omega_{\text{crit}} = \sqrt{\frac{k_{\text{sh}}}{m_e}}$ can amplify or attenuate the transmitted force.

1. **Bearing** (stiffness k_b, damping c_b): The rolling-element

bearing connects inner race to outer race (motor housing). The Hertzian contact stiffness of a typical deep-groove ball bearing is $k_b \sim 10^7$-10^8 N/m, far above the frequencies of interest here, so bearing deflection is negligible.

1. **Motor mount bolts and plate**: The motor housing is assumed

rigidly bolted to the plate. Any compliance in this joint (bolt stretch, washer deformation) is folded into the effective bearing stiffness.

The *bearing transmission function* $T_{b(\omega)}$ quantifies how much of the raw centrifugal force $m_e e \omega^2$ actually reaches the plate at frequency ω:

$$T_{b(\omega)} = \frac{k_b + i c_b \omega}{(k_b - m_e \omega^2) + i c_b \omega} \tag{79}$$

The force entering the plate is therefore:

$$F_{\text{plate}}(\omega) = T_{b(\omega)} \cdot m_e e \omega^2 \tag{80}$$

Rigid-bearing limit. When $k_b \gg m_e \omega^2$ (bearing far from its own resonance):

$$T_{b(\omega)} \to 1 (k_b \to \infty) \tag{81}$$

All force transfers to the plate. Rolling-element bearings typically have Hertzian contact stiffnesses $k_b \sim 10^7$-10^8 N/m. As long as $m_e \omega^2 \ll k_b$ - satisfied at any bistable resonance frequency well below the bearing's own resonance - $T_b \approx 1$ and the full centrifugal force reaches the plate, justifying $F_{\text{motor}} = m_e e \omega^2 \sin(\omega t)$.

Single-Axis Projection of the 2-D Rotating Force

The centrifugal force is a vector that rotates in the plane of the motor shaft:

$$\boldsymbol{F}_{\text{cent}}(t) = m_e e\omega^2[\cos(\omega t)\hat{x} + \sin(\omega t)\hat{y}] \tag{82}$$

Only the component aligned with the bistable stroke direction (take $\hat{y}$ as the snap axis) does useful work on the bistable:

$$F_{\text{snap}}(t) = \boldsymbol{F}_{\text{cent}}(t) \cdot \hat{y} = m_e e\omega^2 \sin(\omega t) \tag{83}$$

The orthogonal component $m_e e\omega^2 \cos(\omega t)$ loads the plate transversely; it is transmitted to the plate support as a lateral force but does not couple into the bistable snap direction and is neglected in the 1-DOF model. **Practical implication.**

Aligning the motor shaft so that the eccentricity plane contains the snap direction maximises the effective force. Misalignment by angle α reduces F_{snap} by $\cos\alpha$.

Motor Shaft Angle Does Not Appear in the Hook Dynamics

A key property of the rotating-unbalance architecture is that the *motor shaft angle* $\theta(t) = \omega t + \theta_0$ does not appear in the bistable hook's equation of motion - only the resulting plate acceleration does. **Proof.**

The hook EOM in relative coordinates is (equation Equation 50):

$$m_h \ddot{x}_{\text{rel}} = F_{\text{restore}}(x_{\text{rel}}) - c_h \dot{x}_{\text{rel}} - m_h \ddot{x}_{P(t)} \tag{84}$$

The right-hand side contains $\ddot{x}_{P(t)}$ - the plate acceleration. This quantity depends on $F_{\text{snap}}(t) = m_e e\omega^2 \sin(\omega t)$ through the plate dynamics, but the shaft angle θ itself never appears. Whether the motor has rotated 100 turns or 1000 turns is irrelevant; only the instantaneous force amplitude and frequency matter. **Contrast with direct drive.**

In a crank-slider or lever-arm direct-drive arrangement, the motor shaft angle θ sets the hook position geometrically: $x_{\text{hook}} = f(\theta)$. A change in θ immediately moves the hook. With rotating unbalance there is no such kinematic constraint: the hook position is determined entirely by the energy balance between the inertial excitation and the bistable restoring force. The motor can spin freely at constant speed without tracking or position control.

Summary of Model Assumptions

p{4.5cm} p{5.0cm}} **Assumption**	**Mathematical form**	**Validity / break-down**
Rigid bearing	$T_{b(\omega)} = 1$	Valid: $k_b \gg m_e\omega^2$ near the current resonance
Rigid motor shaft	$\omega_{\text{crit}} \gg \omega$	Valid if shaft critical speed $\gg f_{\text{n},1}$
Constant motor speed	$\dot{\omega} = 0$	Valid if motor torque bandwidth $\gg f_{\text{n},1}$
Single-axis force	$F = m_e e\omega^2 \sin(\omega t)$	Loses factor $\cos\alpha$ if shaft misaligned
Rigid motor-to-plate joint	$k_{\text{joint}} \to \infty$	Valid for bolted flange; invalid for press-fit
Plate as rigid body	No plate bending modes	Valid if plate structural modes $\gg f_{\text{n},1}$
Linear plate support	k_s, c_s constant	Invalid near plate support nonlinearity

Table 1: Assumptions in the shared-plate rotating-unbalance model and their validity for the Rolly deployment mechanism.

Motor Housing on the Plate with a Proof Mass on the Output Shaft

This subsection treats a distinct configuration: the motor housing is bolted to the bistable plate as before, but instead of a continuously spinning eccentric mass, the motor output shaft drives a *proof mass* M_{pm} that **oscillates linearly** along the snap axis relative to the plate. This is sometimes called a *reaction-mass actuator* or *proof-mass actuator* and is used when independent control of force amplitude and frequency is required.

Equations of Motion

Define coordinates:

- $x_{P(t)}$: absolute plate displacement (positive in snap direction)
- $\delta(t) = x_{\text{pm}}(t) - x_{P(t)}$: proof mass displacement *relative* to plate
- $x_{\text{rel}}(t)$: hook displacement relative to plate (unchanged from equation Equation 50)

The motor applies an internal force $F_{\text{act}}(t)$ between the plate (housing) and the proof mass. By Newton's third law the plate receives $-F_{\text{act}}$ and the proof mass receives $+F_{\text{act}}$. **Proof mass (absolute frame):**

$$M_{\text{pm}}\ddot{x}_{\text{pm}} = M_{\text{pm}}\left(\ddot{x}_P + \ddot{\delta}\right) = F_{\text{act}}(t) \tag{85}$$

Plate (absolute frame):

$$M_P\ddot{x}_P = -F_{\text{act}}(t) - k_s x_P - c_s\dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h\dot{x}_{\text{rel}} \tag{86}$$

Adding equation Equation 85 and equation Equation 86 to eliminate F_{act}:

$$\big(M_P + M_{\text{pm}}\big)\ddot{x}_P + M_{\text{pm}}\ddot{\delta} = -k_s x_P - c_s \dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h \dot{x}_{\text{rel}} \tag{87}$$

The key observation is that F_{act} has been eliminated. The plate equation is now driven entirely by the proof mass *relative acceleration* $\ddot{\delta}$.

Prescribed Harmonic Proof-Mass Motion

If the motor drives the proof mass to oscillate at amplitude Δ and frequency ω:

$$\delta(t) = \Delta \sin(\omega t) \Rightarrow \ddot{\delta}(t) = -\Delta\omega^2 \sin(\omega t) \tag{88}$$

Substituting into equation Equation 87:

$$\big(M_P + M_{\text{pm}}\big)\ddot{x}_P = \underbrace{M_{\text{pm}}\Delta\omega^2 \sin(\omega t)}_{\text{effective driving force}} - k_s x_P - c_s \dot{x}_P - F_{\text{restore}}(x_{\text{rel}}) + c_h \dot{x}_{\text{rel}} \tag{89}$$

This has *exactly the same form* as the rotating-unbalance plate equation (equation Equation 48), with two substitutions:

$$m_e e \to M_{\text{pm}}\Delta \text{ (force product)} \tag{90}$$

$$M_P \to M_P + M_{\text{pm}} \text{ (effective plate mass)} \tag{91}$$

All plate-transfer-function and comparison results carry over with these substitutions.

Snap Condition for the Proof-Mass Configuration

Applying the same derivation as the tuned-plate case with substitutions equation Equation 90 and equation Equation 91: **Tuned plate ($\boldsymbol{f_P = f_{\text{n},1}}$):**

$$M_{\text{pm}}\Delta \geq 4\zeta_h \zeta_P \big(M_P + M_{\text{pm}}\big) d_s \tag{92}$$

Defining the proof-mass ratio $\mu = \frac{M_{\text{pm}}}{M_P}$ and rearranging:

$$\Delta \geq 4\zeta_h \zeta_P M_P d_s \frac{1+\mu}{\mu} = \frac{4\zeta_h \zeta_P M_P d_s}{\mu} + 4\zeta_h \zeta_P d_s \tag{93}$$

As $\mu \to \infty$ (proof mass much heavier than plate), the required stroke $\Delta \to 4\zeta_h \zeta_P d_s$, independent of M_P. As $\mu \to 0$ (very light proof mass), $\Delta \to \infty$: a tiny mass must oscillate with enormous stroke to produce enough force.

Comparison with Rotating Unbalance

p{4.8cm} p{5.2cm}} **Property**	**Rotating unbalance**	**Proof-mass oscillator**
Force amplitude	$m_e e\omega^2$ (fixed by geometry)	$M_{\text{pm}}\Delta\omega^2$ (Δ controllable)
Force-frequency	Always grows as ω^2	Grows as ω^2 unless $\Delta(\omega)$ is adjusted
Effective plate mass	M_P	$M_P + M_{\text{pm}}$
Motor direction	Continuous rotation (one direction)	Reciprocating (reverses each half-cycle)
Snap product	$m_e e \geq 4\zeta_h\zeta_P M_P d_s$	$M_{\text{pm}}\Delta \geq 4\zeta_h\zeta_{P(M_P+M_{\text{pm}})}d_s$
Shaft angle in EOM?	No	No (same decoupling argument)

Table 2: Proof-mass oscillator vs. rotating unbalance: key differences for the same bistable plate.

The proof-mass oscillator is strictly more flexible than the rotating unbalance because Δ can be varied in real time via motor current. The cost is that the motor must reverse direction each cycle rather than spinning continuously, and the heavier effective plate mass $\left(M_P + M_{\text{pm}}\right)$ raises the snap threshold.

Proof-Mass Resonance

If the shaft connecting motor to proof mass has compliance k_{pm}, the proof mass forms a secondary resonance:

$$\omega_{\text{pm}} = \sqrt{\frac{k_{\text{pm}}}{M_{\text{pm}}}} \tag{94}$$

Driving near ω_{pm} amplifies Δ by $Q_{\text{pm}} = \frac{1}{2\zeta_{\text{pm}}}$, reducing the required motor force by the same factor. This creates a *triple resonance* path: plate resonance at ω_P, bistable hook resonance at $\omega_{\text{n},1}$, and proof-mass resonance at ω_{pm}. Tuning all three to the same frequency gives amplification $Q_P \cdot Q_h \cdot Q_{\text{pm}}$ but also narrows the bandwidth and makes the system sensitive to parameter variation.

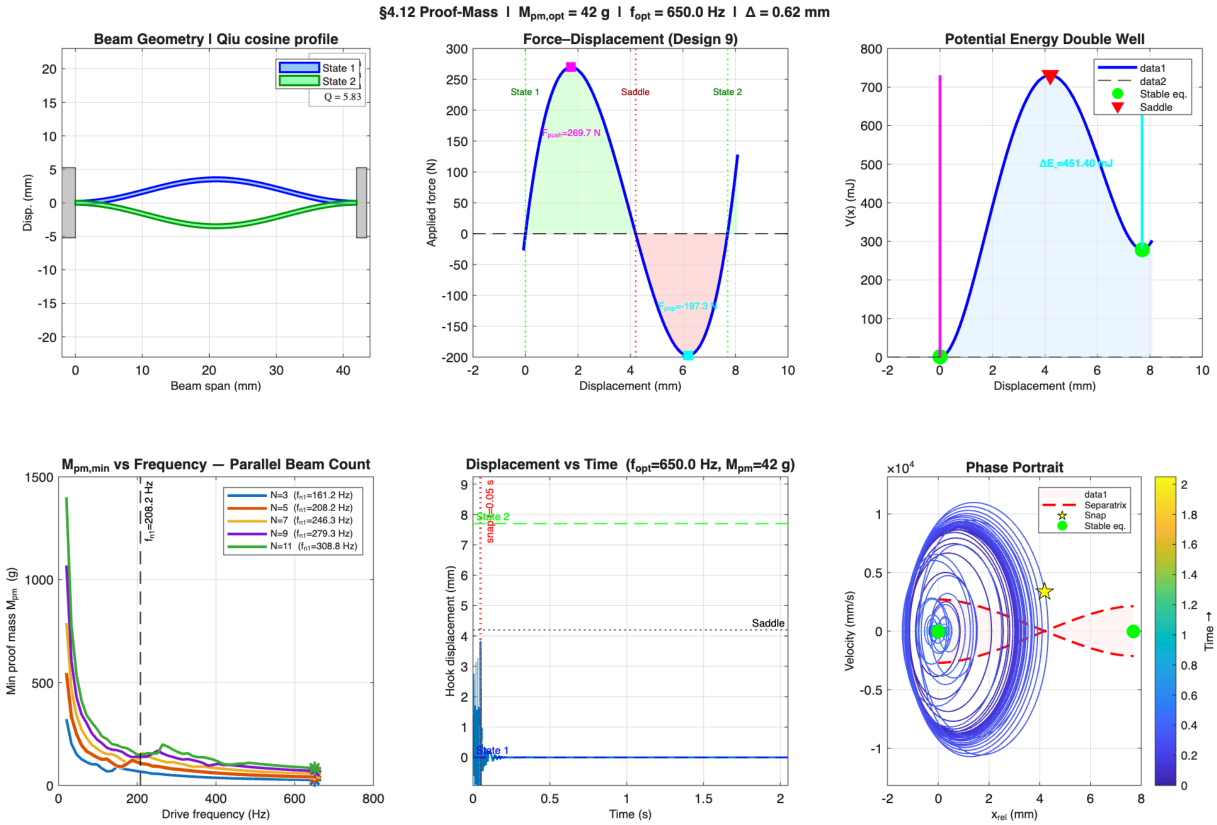

Figure 3: Proof-mass actuation analysis: minimum proof mass $M_{\text{pm,min}}$ and corresponding snap-through drive frequency as a function of design parameters (N parallel beams, hook mass m_h).

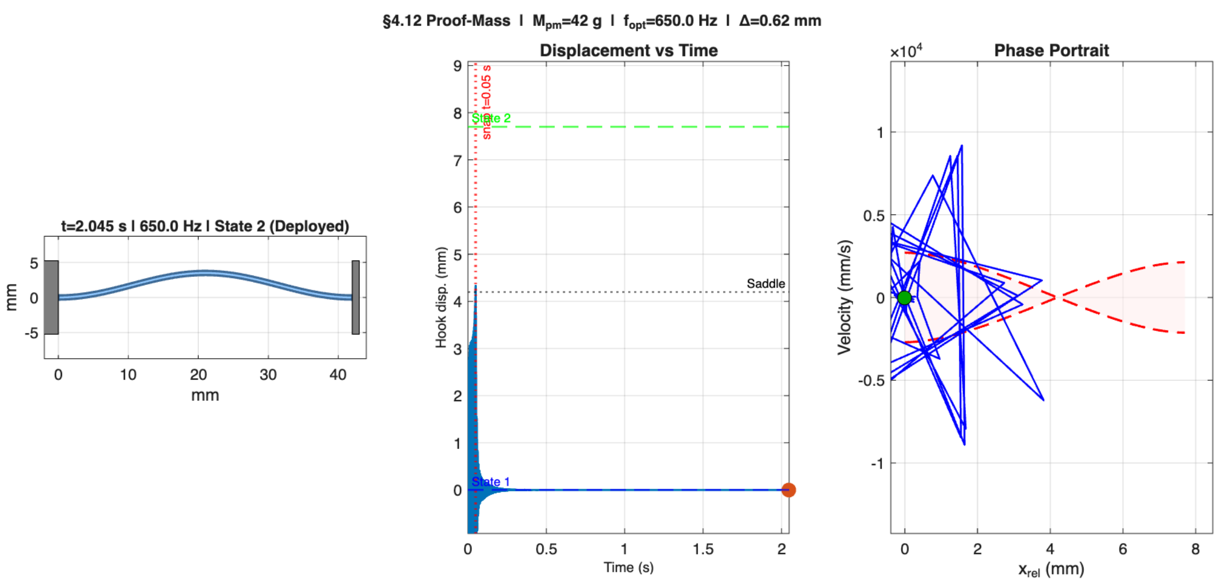

Figure 4: Animation snapshot for the proof-mass actuator at the optimal proof-mass parameter. The proof mass oscillates on the motor output shaft, driving the plate and hook into snap-through.

Actuator Limit Used in the Simulation

The detailed RS05 motor specifications and CAN notes have been moved to `rs05_motor_specifications.tex` and `rs05_motor_specifications.typ`. The main model only needs the shaft velocity limit used by the optimization.

The RS05 no-load speed is treated as a shaft angular-velocity limit:

$$\omega_{\text{max}} = 50.3 \text{ rad/s} \tag{95}$$

For sinusoidal oscillation at frequency f with angular shaft amplitude Θ:

$$v_{\text{peak}} = \Theta \cdot 2\pi f \leq \omega_{\text{max}} \tag{96}$$

Thus the maximum angular amplitude is:

$$\Theta_{\text{max}}(f) = \frac{\omega_{\text{max}}}{2\pi f} \tag{97}$$

Through a lever arm r, the corresponding linear stroke is:

$$\delta_{\text{max}}(f) = \frac{\omega_{\text{max}} r}{2\pi f} \tag{98}$$

For snap-through, the available stroke must reach the saddle. The minimum lever arm is therefore:

$$r_{\text{min}}(f) = \frac{2\pi f d_s}{\omega_{\text{max}}} \tag{99}$$

Optimization

Snap-Through Threshold

From linear resonance theory, the steady-state hook amplitude at $\omega = \omega_{\mathrm{n},1}$ is:

$$A_{\mathrm{ss}} = \frac{F_0}{2\zeta_h k_{\mathrm{eff},1}} \tag{100}$$

Snap-through occurs when $A_{\mathrm{ss}} \geq d_s$, giving:

$$F_{0,\mathrm{min}} = 2\zeta_h k_{\mathrm{eff},1} d_s \tag{101}$$

This is the *minimum force applied directly at the hook*. For base excitation (motor on plate), the motor force required is larger by a factor that depends on the plate dynamics and drive frequency.

Mass Actuation Force

The central design question for the Rolly deployment mechanism is: *for a given mass budget added to the robot, what is the maximum force that mass can deliver to the bistable hook?*

Proof-Mass Oscillator (proof-mass configuration)

A proof mass M_{pm} oscillating at frequency ω and amplitude Δ delivers inertial force $F = M_{\mathrm{pm}}\Delta\omega^2$. At the velocity limit (equation Equation 97):

$$\Delta_{\mathrm{max}}(\omega) = \frac{\omega_{\mathrm{max}} r}{\omega} \tag{102}$$

The maximum force per unit proof mass is:

$$\frac{F_{\mathrm{max}}}{M_{\mathrm{pm}}} = \Delta_{\mathrm{max}}\omega^2 = \omega_{\mathrm{max}} r \omega \tag{103}$$

This grows *linearly with drive frequency*. Higher frequency delivers more force for the same proof mass — up to the limit imposed by the bistable dynamics (TMD anti-resonance at $\omega_{\mathrm{n},1}$; see the tuned-plate case). The coupled relative-mode resonance at ω_{rel} is the correct drive frequency:

$$\omega_{\mathrm{rel}} = \sqrt{\frac{k_{\mathrm{eff},1}}{M_{\mathrm{red}}}}, M_{\mathrm{red}} = \frac{m_h M_P}{m_h + M_P} \tag{104}$$

The minimum proof mass to snap at ω_{rel} is found by binary search over the nonlinear ODE.

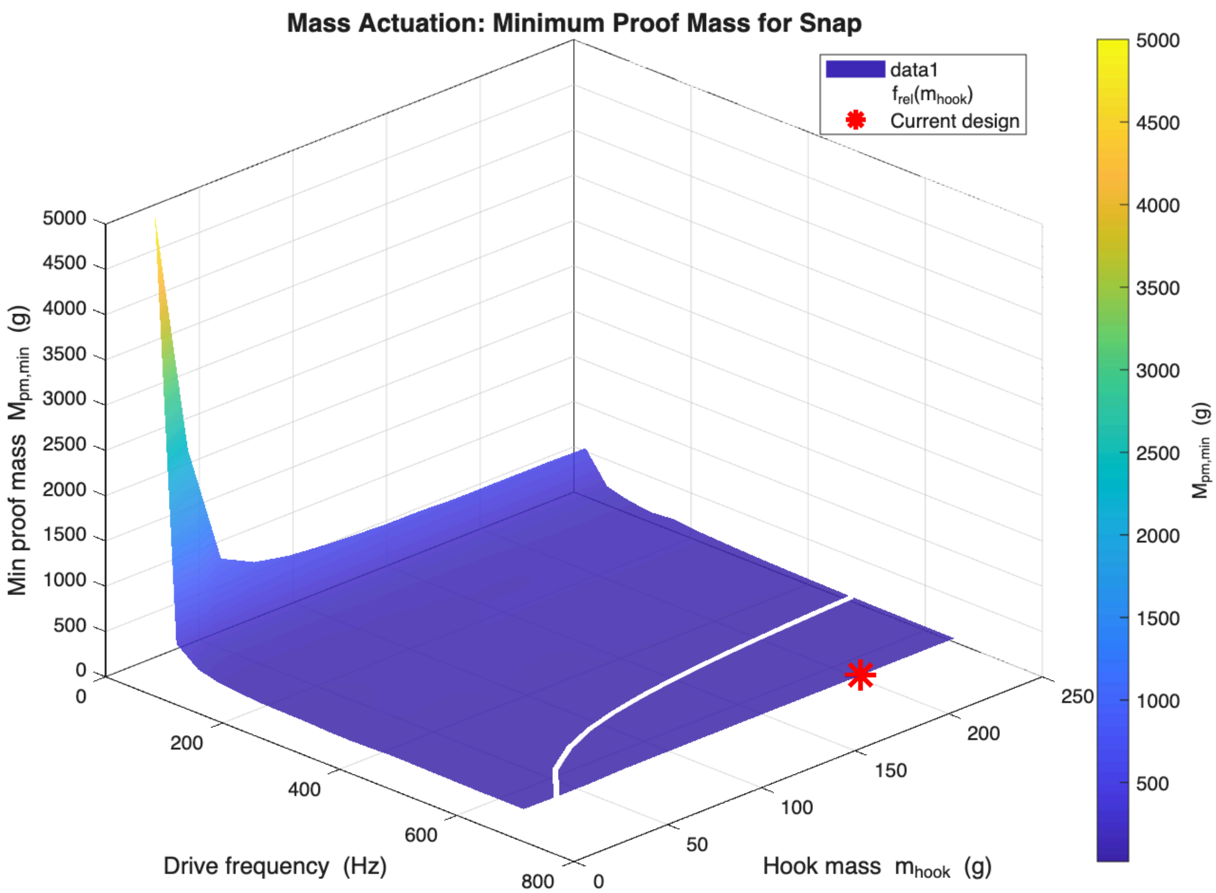

Figure 5: $M_{\mathrm{pm,min}}$ vs. drive frequency and hook mass m_h.

Config 2: Prescribed Base Motion (no added mass)

The motor prescribes the plate displacement $x_{P(t)} = A \sin(\omega_{\mathrm{n},1} t)$. The minimum plate amplitude for resonant snap is:

$$A_{\mathrm{min}} = 2\zeta_h d_s \tag{105}$$

The required motor torque at resonance ($\omega = \omega_{\mathrm{n},1}$):

$$\tau = m_h A_{\mathrm{min}} \omega_{\mathrm{n},1}^2 r = 2\zeta_h k_{\mathrm{eff},1} d_s r \tag{106}$$

This is *independent of hook mass* m_h. No additional hardware mass is required - the motor already present for locomotion is used.

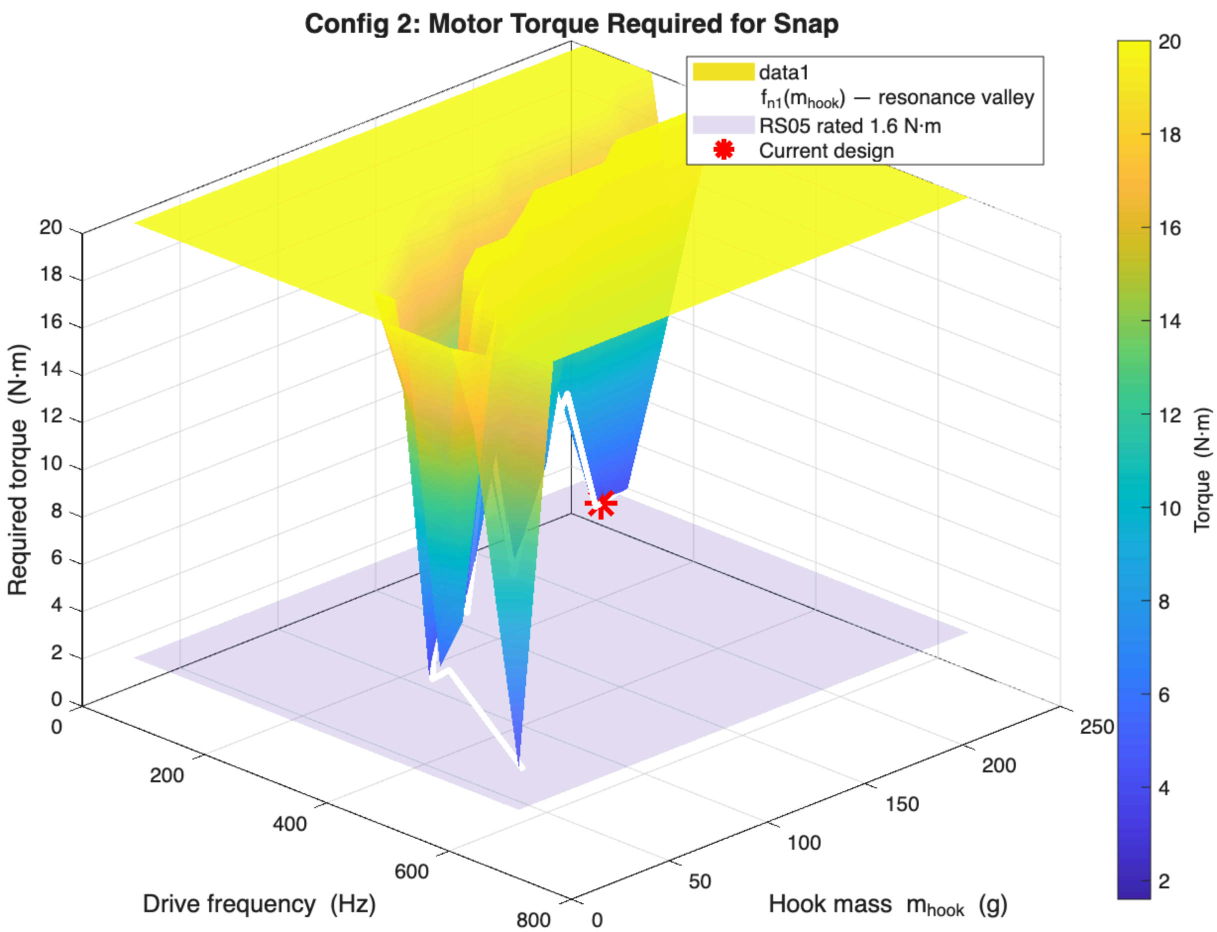

Figure 6: Required torque vs. drive frequency and hook mass m_h.

Parallel Beam-Count Scaling (N beams)

For the current model, N is the number of parallel bistable beams. The beams share the same hook displacement, so adding beams increases force and stiffness, not stroke:

$$d_f = d_{\mathrm{f},1} \text{ (stroke unchanged)} \tag{107}$$

$$d_s = d_{\mathrm{s},1} \text{ (saddle unchanged)} \tag{108}$$

$$k_{\mathrm{eff},1} \propto N \text{ (stiffness rises)} \tag{109}$$

$$f_{\mathrm{n},1} \propto \sqrt{N} \text{ (natural frequency rises)} \tag{110}$$

$$F_{0,\min} \propto N \text{ (force threshold rises)} \tag{111}$$

The old series assumption made the saddle displacement look too large. In the parallel model, $N = 5$ raises the stored energy and stiffness while keeping the hook travel at the single-beam value. The post-snap potential energy available to accelerate the hook to State 2:

$$\Delta E_2 = V(d_s) - V\left(d_f\right) \tag{112}$$

grows with N because the force-displacement curve scales upward, not because the stroke grows. This is the *impact energy* available when the hook collides with a target object at State 2.

Impact Force at State 2

When the hook arrives at $x_{\text{rel}} = d_f$ it has kinetic energy accumulated from two sources:
1. **Kinetic energy at saddle crossing** KE_{saddle}

(from resonant build-up).
1. **Potential energy released** by the bistable spring from saddle

to State 2: ΔE_2.

The hook velocity at impact (ignoring damping over the short snap transit):

$$v_{\text{impact}} = \sqrt{v_{\text{saddle}}^2 + \frac{2\Delta E_2}{m_h}} \tag{113}$$

At the minimum snap threshold $v_{\text{saddle}} \approx 0$, so $v_{\text{impact}} \approx \sqrt{2\Delta \frac{E_2}{m_h}}$. The peak force on the target object (Hertz contact model):

$$F_{\text{peak}} = v_{\text{impact}} \sqrt{k_{\text{contact}} m_h} \tag{114}$$

where k_{contact} is the contact stiffness. The impact force scales as $\sqrt{k_{\text{contact}}}$: steel-on-steel ($k \sim 10^{10}$ N/m) gives $\sim 3000 \times$ more peak force than rubber-on-rubber ($k \sim 10^4$ N/m) for the same impact velocity. The impact momentum (relevant for deforming targets):

$$p = m_h v_{\text{impact}} = \sqrt{2 m_h \Delta E_2} \tag{115}$$

grows as $\sqrt{m_h}$ - heavier hooks deliver more momentum.

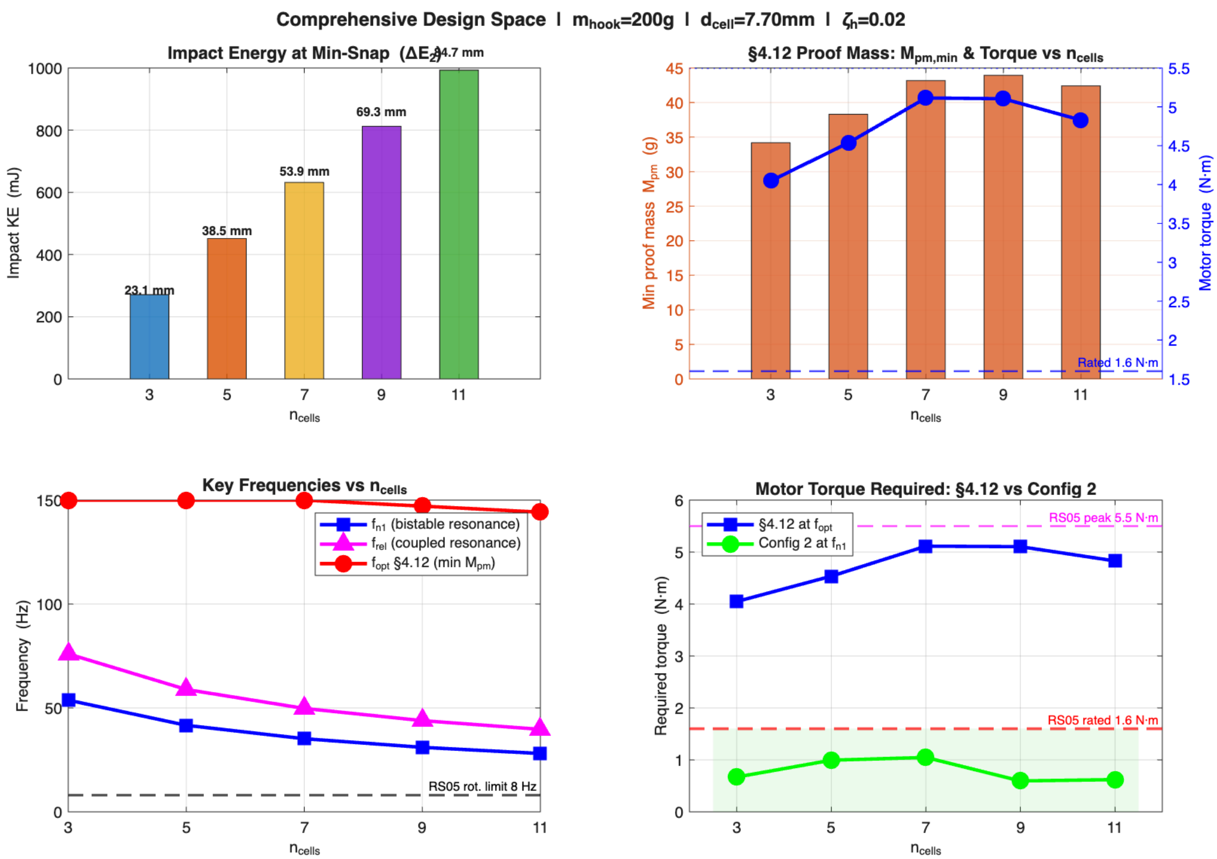

Figure 7: Comprehensive design space: key performance metrics (natural frequency $f_{\mathrm{n},1}$, energy barrier ΔE_1, impact energy ΔE_2, minimum required force $F_{0,\min}$, and minimum proof mass $M_{\mathrm{pm,min}}$) as a function of the number of parallel beams N. All curves are normalised to their single-beam values.

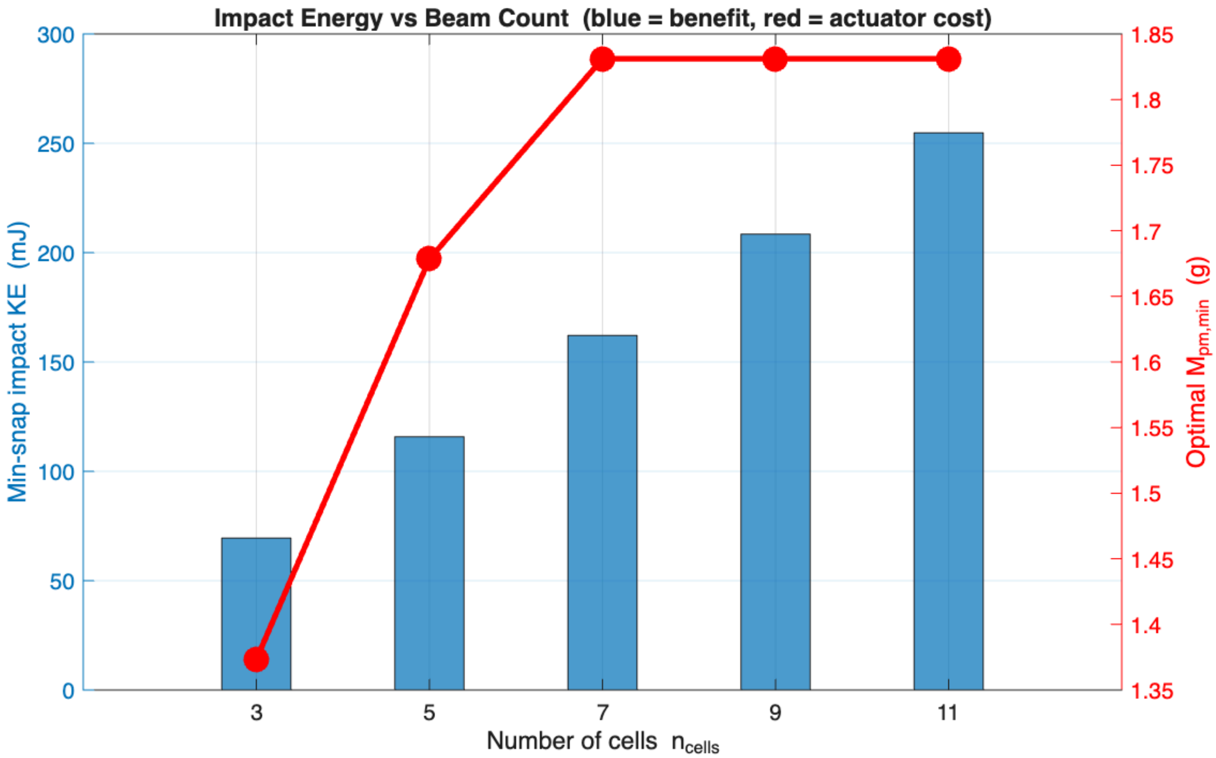

Figure 8: Impact energy ΔE_2 and actuation cost metrics as a function of the number of parallel beams N. Impact energy grows with N because the force-displacement curve scales upward.

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